# MTH105

**Timed Online Assignment – January Semester 2022**

## Fundamentals of Mathematics

**Saturday, 21 May 2022** **1:00 pm – 3:00 pm**

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**Time allowed: 2 hours**

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**INSTRUCTIONS TO STUDENTS:**

1. This Timed Online Assignment (TOA) contains **SEVEN (7)** questions and comprises **SIX (6)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

*Answer all questions. (Total 100 marks)*

**Question 1**

(a) Employ a truth table to determine if the following argument is valid or invalid. Indicate clearly which rows of the truth table are critical rows.

| | |
|---|---|
| $\sim p \vee q$ | (premise) |
| $p \longrightarrow (q \wedge r)$ | (premise) |
| $\sim r$ | (premise) |
| $\therefore \quad q$ | (conclusion) |

(7 marks)

(b) Construct a chain of logical equivalences to show that

$$\sim\big((p \longrightarrow q) \wedge (r \longrightarrow q)\big) \equiv (p \vee r) \wedge \sim q.$$

Do not use truth tables in this part of the question.

(5 marks)

(c) Use the Rules of Inference to prove that the following argument form is valid.

| | |
|---|---|
| $p \wedge (q \vee r)$ | (premise) |
| $\sim s \longrightarrow \sim r$ | (premise) |
| $\therefore \quad p \wedge (q \vee s)$ | (conclusion) |

Do not use truth tables in this part of the question.

(8 marks)

**Question 2**

(a) Give a counter-example to show that the following statement is false.

$$\forall x \in \mathbb{N}\ \forall y \in \mathbb{N}\ \ (x^3 + y^3 \text{ is divisible by } 7) \longrightarrow (x^2 + y^2 \text{ is divisible by } 7)$$

(3 marks)

(b) Provide the negation of the statement

$$\forall x \in \mathbb{R}\ \exists y \in \mathbb{Z}\ \exists z \in \mathbb{Q}\ \ (x + y + z > xyz) \wedge (z > 1).$$

(3 marks)

(c) Let

$$D = \left\{ x \in \mathbb{Q}^+ \ : \ \exists\, m, n \in \mathbb{N}, m \text{ is odd}, x = \frac{m}{2^n} \right\}.$$

(i) Given that the variable $y$ has domain $D$, determine the truth set of the predicate

$$\left( \left( \frac{1}{2} < y < \frac{15}{16} \right) \vee \left( \frac{33}{32} < y < \frac{3}{2} \right) \right) \wedge (16y \in \mathbb{N}).$$

(4 marks)

(ii) Employ a proof by contradiction to show that there are an infinite number of elements in $D$ that are between 0 and 1.

(4 marks)

(iii) Use the result of Question 2(c)(ii) to show that whenever $p, q$ are positive integers with $p < q$, there are an infinite number of elements in $D$ that are between $p$ and $q$.

(4 marks)

**Question 3**

(a) Employ either a proof by contraposition or a proof by contradiction to show that if $n, m, p, q$ are positive integers satisfying

$$\frac{13n + 11m}{2p + q} \geq 7\,,$$

then either $n \geq 9$, or $m \geq 11$, or $p \leq 8$, or $q \leq 12$.

(8 marks)

(b) Show that for all integers $n$,

$$\gcd(n^6 + 3n^2 + n + 4\,, n^2 + 1) = 1.$$

(7 marks)

**Question 4**

Use mathematical induction to prove that

$$\sum_{r=1}^{n} \frac{1}{(r+2)(r+3)(r+4)} = \frac{n(n+7)}{24(n+3)(n+4)}$$

for all positive integers $n$.

(10 marks)

**Question 5**

(a) Use the Euclidean Algorithm to find $d = \gcd(2676,360)$ and express $d$ as an integral linear combination of $2676$ and $360$.

(8 marks)

(b) Define the sets

$$A = \{1,2,3,6,8,10\}, B = \{2,4,6,7,10,11\}, C = \{2,3,4\}.$$

List the elements of the sets $(A \cap B) \cup C$, $(A \setminus B) \times C$, and $\mathcal{P}(C)$, where $\mathcal{P}(C)$ denotes the power set of $C$.

(7 marks)

**Question 6**

Define a function $f: \mathbb{N} \times \mathbb{N} \longrightarrow \mathbb{Q}$,

$$f\big((a,b)\big) = a + \frac{1}{b}.$$

(a) Show that $f$ is not surjective.

(3 marks)

(b) Determine with justification whether or not $f$ is injective.

(7 marks)

**Question 7**

It is given that $A, B, C$ are events that satisfy the following:

$P(A) = 3/10$
$P(B) = 59/100$
$P(C) = 1/3$
$P(B \cap C) = 7/60$
$P(A \cap B \cap C) = 1/15$

It is also given that $A, C$ are independent events.

(a) Find $P(A \triangle C)$, where $A \triangle C$ denotes the symmetric difference of $A$ and $C$. In your working, you may use without justification the fact that $A \triangle C = (A \cup C) \setminus (A \cap C)$.

(4 marks)

(b) Show using a rigorous mathematical argument that

$$\frac{1}{150} \leq P\left(A \cap B \cap \overline{C}\right) \leq \frac{1}{5},$$

where $\overline{C}$ denotes the complement of $C$.

(8 marks)

**----- END OF PAPER -----**